FED. SUP CLASS
5305

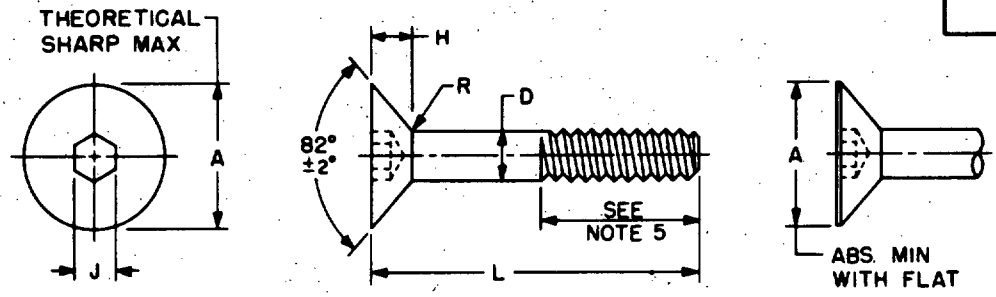


TABLE I

	NOMINAL SIZE	.112	.138	.164	.190	.250	.3125	.375	.4375	.500	.625	.750
	THREADS PER INCH, UNC-3A	40	32	32	24	20	18	16	14	13	11	10
D	BODY DIAMETER MAX	.1120	.1380	.1640	.1900	.2500	.3125	.3750	.4375	.5000	.6250	.7500
	BODY DIAMETER MIN	.1075	.1329	.1585	.1840	.2435	.3053	.3678	.4294	.4919	.6163	.7406
A	HEAD DIAMETER SHARP MAX	.255	.307	.359	.411	.531	.656	.781	.844	.937	1.188	1.438
	HEAD DIAMETER ABS MIN	.218	.263	.311	.359	.480	.600	.720	.781	.872	1.112	1.355
H	HEAD HEIGHT REF	.083	.097	.112	.127	.161	.198	.234	.234	.251	.324	.396
J	SOCKET SIZE NOM	.062	.078	.094	.125	.156	.188	.219	.250	.312	.375	.500
R	RADIUS Ⓒ MAX	.012	.015	.015	.015	.015	.015	.015	.015	.015	.015	.015
	TENSILE STRENGTH, LBS-MIN	480	725	1,120	1,400	2,540	4,190	6,200	8,500	11,300	18,000	23,300

L LENGTH	TOLERANCE Ⓒ SIZE .112 THRU .375	TOLERANCE Ⓒ OVER SIZE .375	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
.250			1	7									
.375			2	8	13	19	27	37					
.500			3	9	14	20	28	38	49				
.625	+.000 -.030	+.000 -.030	4	10	15	21	29	39	50	110*			
.750			5	11	16	22	30	40	51	62*	73		
.875										111*	112*	113*	
1.000					17	23	31	41	52	63*	74	114*	115*
1.250						24	32	42	53	64*	75	84	93
1.500						25	33	43	54	65*	76	85	94
1.750	+.000 -.040	+.000 -.060						44	55	66*	77	86	95
2.000								45	56	67*	78	87	96
2.250								46	57	68*	79	88	97
2.500								47	58	69*	80	89	98
2.750	+.000 -.060	+.000 -.080							59	70*	81	90	99
3.000									60	71*	82	91	100

Ⓒ * INACTIVE FOR NEW DESIGN AFTER 20 February 1974.

NOTES:

1. MATERIAL: Steel, corrosion-resisting, in accordance with Fed. Std. No. 66; AISI 302, 303, 304, 305, 316 or 384 (developed for cold heading). Minimum ultimate tensile strength shall be 80,000 PSI for sizes thru 0.625 and 70,000 PSI for sizes over 0.625.
2. PROTECTIVE COATING: Passivated in accordance with QQ-P-35.
3. MAGNETIC PERMEABILITY: These screws shall have a magnetic permeability of 2.0 maximum (air = 1.0) for a field strength H = 200 oersteds when using a magnetic indicator in accordance with MIL-I-17214.
4. THREADS: Threads shall be in accordance with the procurement specification.
5. THREAD LENGTH: Minimum full form thread length shall be equal to 2D plus 0.500 inch. The tolerance shall be equal to the nominal size. Screws too short for application of the formula shall be threaded as close to the head as practicable.
6. DIMENSIONS: All dimensions in Table I are in inches. All dimensions in Table II are in millimetres.
7. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS24671-1.
8. Referenced documents shall be of the issue in effect on date of invitations for bid.
9. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

Ⓒ For changes see sheets 1 and 2.

REVIEWER: AV, IS, ME
USER: AS, AT, MC, MU, OS, SH, WV

APPROVED 6 APR 1960 REVISED (A) 19 DEC 1963 (B) 20 JUL 1966 Ⓒ 20 FEB 1974

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC Other Cust 82	TITLE SCREW, CAP, SOCKET-HEAD-FLAT COUNTERSUNK, 82°, CORROSION-RESISTING STEEL, UNC-3A	MILITARY STANDARD MS 24671
PROCUREMENT SPECIFICATION FF-S-86	SUPERSEDES: MS35249 in part	SHEET 1 OF 2

DD FORM 1 SEP 67 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:59Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS 5305

REVIEWER: AV, IS, ME
USER: AS, AT, MC, MU, OS, SH, WV

TABLE II
METRIC TRANSLATION OF TABLE I

	NOMINAL SIZE		2.84	3.50	4.17	4.83	6.35	7.94	9.52	11.11	12.70	15.88	19.05
	PITCH		.64	.79	.79	1.06	1.27	1.41	1.59	1.81	1.95	2.31	2.54
D	BODY DIAMETER	MAX	2.84	3.50	4.17	4.83	6.35	7.94	9.52	11.11	12.70	15.88	19.05
		MIN	2.73	3.38	4.03	4.67	6.18	7.75	9.34	10.91	12.49	15.65	18.81
A	HEAD DIAMETER	SHARP MAX	6.48	7.80	9.12	10.44	13.49	16.66	19.84	21.44	23.80	30.18	36.53
		ABS MIN	5.54	6.68	7.90	9.12	12.19	15.24	18.29	19.84	22.15	28.24	34.42
H	HEAD HEIGHT	REF	2.11	2.46	2.84	3.23	4.09	5.03	5.94	5.94	6.38	8.23	10.06
J	SOCKET SIZE	NOM	1.57	1.98	2.39	3.18	3.96	4.78	5.56	6.35	7.92	9.52	12.70
R	RADIUS	MAX	.30	.38	.38	.38	.38	.38	.38	.38	.38	.38	.38

L LENGTH	TOLERANCE: SIZE 2.84 THRU 9.52	TOLERANCE: OVER SIZE 9.52	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
6.35			1	7									
9.52			2	8	13	19	27	37					
12.70			3	9	14	20	28	38	49				
15.88	+.00 -.76	+.00 -.76	4	10	15	21	29	39	50	110*			
19.05			5	11	16	22	30	40	51	62*	73		
22.22										111*	112*	113*	
25.40					17	23	31	41	52	63*	74*	114*	115*
31.75						24	32	42	53	64*	75	84	93
38.10						25	33	43	54	65*	76	85	94
44.45	+.00 -1.02	+.00 -1.52						44	55	66*	77	86	95
50.80								45	56	67*	78	87	96
57.15								46	57	68*	79	88	97
63.50								47	58	69*	80	89	98
69.85	+.00 -1.52	+.00 -2.03							59	70*	81	90	99
76.20									60	71*	82	91	100

* INACTIVE FOR NEW DESIGN AFTER 20 February 1974.

INTERCHANGEABILITY

Screws covered by the following dash numbers given in MS35249 in part, are canceled after 19 Dec 1963. The canceled MS35249 screws cannot always replace the MS24671 screws and should be used until existing stocks are depleted. Use only the MS24671 screws for new design and replacement. Replacement shall be in accordance with Table III.

TABLE III

PART NUMBER		PART NUMBER		PART NUMBER		PART NUMBER	
CANCELED MS35249	SUPERSEDED BY MS24671	CANCELED MS35249	SUPERSEDED BY MS24671	CANCELED MS35249	SUPERSEDED BY MS24671	CANCELED MS35249	SUPERSEDED BY MS24671
127	110	140	112	153	NR	166	88
128	62	141	74	154	NR	167	89
129	111	142	75	155	NR	168	90
130	63	143	76	156	NR	169	91
131	64	144	77	157	NR	170	115
132	65	145	78	158	NR	171	93
133	66	146	79	159	NR	172	94
134	67	147	80	160	113	173	95
135	68	148	81	161	114	174	96
136	69	149	82	162	84	175	97
137	70	150	NR	163	85	176	98
138	71	151	NR	164	86	177	99
139	73	152	NR	165	87	178	100

NR = No replacement

APPROVED 6 APR 1960 REVISED (A) 19 DEC 1963 (B) 20 JUL 19 FOR CHANGES SEE SHEETS 1 AND 2.

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC
Other Cust 82

TITLE: SCREW, CAP, SOCKET-HEAD-FLAT COUNTERSUNK, 82, CORROSION-RESISTING STEEL, UNC-3A

MILITARY STANDARD
MS24671

PROCUREMENT SPECIFICATION: FF-S-86

SUPERSEDES: MS35249 in part

SHEET 2 OF 2

DD FORM 1 SEP 67 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:59Z
Check the source to verify that this is the current version before use.